

Nonstandard Educational Careers and the Returns to Vocational Education - Supplemental Appendix

November 28, 2025

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S1 Additional Figures

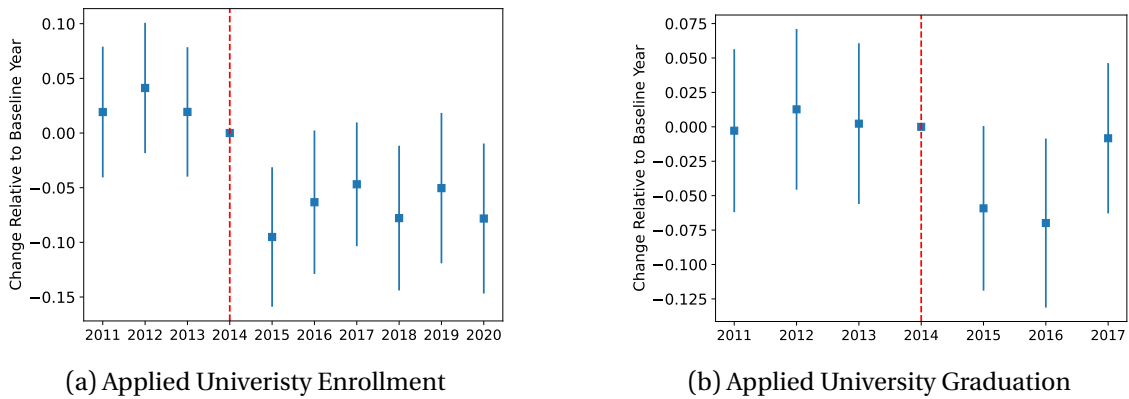


Figure S1: Enrollment and Graduation Effects of the Reform to Student Income Subsidies on affected Students

Note: These figures show the results of a difference-in-differences estimation of the effect of the 2015 reform to student income subsidies on applied university enrollment and graduation. The analysis uses only people who have graduated from a higher vocational program, as these were the only people who showed a decline in enrollment rates following the reform. The treatment group consists of people from low-income households who were likely to move out before the reform. These individuals were most affected by the reform. I use a gradient boosting regressor to predict the probability of moving out in the absence of the reform based on observable characteristics. The treatment group is defined as people who had a more than 70% likelihood of moving out, and the control group as people who had a lower than 30% probability of moving out. The figures show the coefficients of an event study regression where the treatment group is interacted with year dummies. The first figure shows the results for applied university enrollment, and the second figure shows the results for applied university graduation. All of the coefficients are relative to the year 2014, which is the last year before the reform took effect.

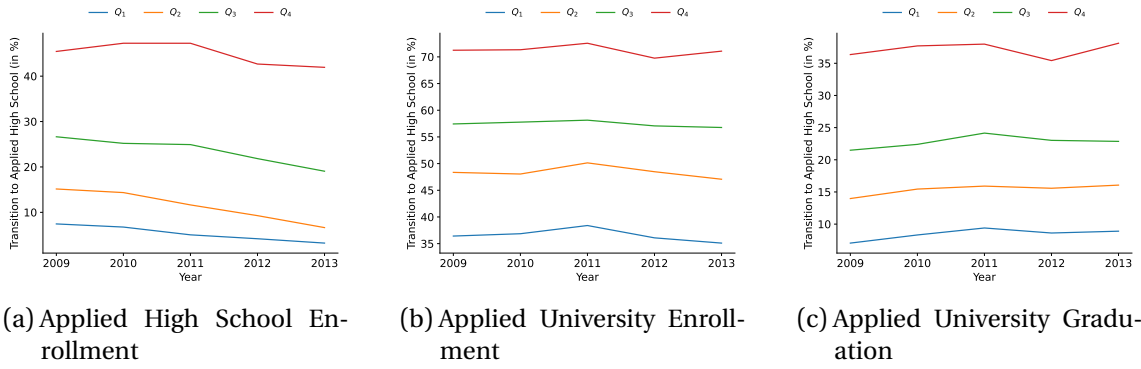


Figure S2: Enrollment and Graduation Rates Between 2009 and 2013

Note: This figure shows enrollment and graduation at applied high school and applied university for cohorts graduating from vocational school between 2009 and 2013 for each quartile of grades at the end of vocational school.

S2 Parameter estimates

In the tables below I use the original abbreviations to refer to different schooling programs. HAVO refers to applied high school, HBO to applied university, MBO3 and MBO4 to different levels of vocational education, and VMBO to pre-vocational secondary education.

Table S1: Nonpecuniary returns to education

	HAVO	HBO	MBO4	MBO3
Constant	-128442 (102) 25479	16108 (90)	56254 (99)	66309 (83)
G_2	(108) 71296	—	—	—
G_3	(92) 132759	—	—	—
G_4	(109) 8598	—	—	—
θ_2	(91) -8928	34150 (91)	-43706 (87)	-37411 (76)
θ_3	(104) -4773	(103) 38325	(100) 27832	(97) 50545
θ_4	(767) 4402	(96) 7115	(90) 7169	(859) -6911
θ_5	(844) 1960	(94) 10343	(110) 3499	(127) -19886
Y_2	(114) -4792	(89) 4389	(84) -1063	(106) -2779
Y_3	(117) -1293	(103) 11345	(78) 462	(83) -29778
Y_4	(876) (876)	(79) (79)	(84) (84)	(101) (101)

Table S2: Nonpecuniary returns to working

	Academic	Vocational
Age	-1157 (90)	3038 (90)
Constant	118755 (93)	33900 (95)
Y_2	14475 (106)	14156 (95)
Y_3	17893 (84)	18354 (100)
Y_4	35539 (85)	30463 (95)
MBO3	—	27171 (82)
MBO4	—	38503 (89)
VMBO	—	-3883 (72)

Table S3: Wage returns

	Academic	Vocational
	0.033	0.018
Age	(0.009)	(0.011)
	-0.002	-0.008
Age $\times I[Age > 30]$	(0.004)	(0.005)
	2.571	2.650
Constant	(0.035)	(0.036)
	0.056	0.027
Experience	(0.013)	(0.009)
	-0.177	-0.118
Experience ²	(0.033)	(0.029)
	0.017	0.031
Experience $\times \theta_2$	(0.011)	(0.007)
	0.038	0.025
Experience $\times \theta_3$	(0.016)	(0.007)
	0.001	0.007
Experience $\times \theta_4$	(0.014)	(0.010)
	0.010	-0.002
Experience $\times \theta_5$	(0.030)	(0.024)
	0.015	0.018
G_2	(0.018)	(0.013)
	0.017	0.016
G_3	(0.017)	(0.013)
	0.004	0.006
G_4	(0.016)	(0.023)
	0.141	-0.038
θ_2	(0.041)	(0.047)
	-0.018	-0.073
θ_3	(0.037)	(0.034)
	0.135	0.095
θ_4	(0.043)	(0.039)
	0.019	0.018
θ_5	(0.042)	(0.047)
		-0.009
Experience $\times$ MBO3	—	(0.008)
		-0.002
Experience $\times$ MBO4	—	(0.007)
		-0.004
Experience $\times$ VMBO	—	(0.010)
		0.035
Experience ² $\times$ MBO3	—	(0.041)
		0.027
Experience ² $\times$ MBO4	—	(0.030)
		0.010
Experience ² $\times$ VMBO	—	(0.039)
		0.151
MBO3	—	(0.028)
		0.164
MBO4	—	(0.031)
		0.097
Dropout	—	(0.032)
		0.029
VMBO	—	(0.034)

Table S4: Dropout Risk

	HBO	MBO4	MBO3	HAVO
Constant	0.436 (0.036)	0.891 (0.041)	0.921 (0.038)	0.262 (0.037)
G_2	0.224 (0.034)	0.135 (0.049)	—	0.323 (0.049)
G_3	0.490 (0.039)	0.184 (0.041)	—	0.610 (0.044)
G_4	0.958 (0.042)	0.217 (0.034)	—	0.906 (0.042)
θ_2	-0.046 (0.047)	—	—	-0.030 (0.053)
θ_3	-0.167 (0.048)	—	—	0.190 (0.049)
θ_4	-0.015 (0.051)	—	—	0.229 (0.042)
θ_5	-0.008 (0.043)	—	—	0.022 (0.052)
Y_2	0.113 (0.037)	0.185 (0.044)	0.051 (0.035)	0.034 (0.043)
Y_3	0.158 (0.039)	0.397 (0.048)	0.316 (0.045)	-0.014 (0.043)
Y_4	0.256 (0.036)	0.356 (0.041)	0.274 (0.040)	-0.008 (0.049)
MBO4	-0.251 (0.037)	—	—	—

Table S5: Degree Duration

	HBO	MBO4	MBO3
Constant	3.897 (0.045)	2.437 (0.043)	1.279 (0.046)
Y_2	-0.082 (0.053)	-0.047 (0.038)	-0.179 (0.050)
Y_3	-0.227 (0.040)	-0.011 (0.041)	-0.192 (0.062)
Y_4	-0.126 (0.042)	-0.026 (0.051)	-0.071 (0.044)
G_2	0.003 (0.045)	-0.121 (0.052)	—
G_3	-0.003 (0.048)	0.006 (0.035)	—
G_4	-0.195 (0.049)	-0.334 (0.050)	—

Table S6: Type probabilities

	Type 2	Type 3	Type 4	Type 5
Constant	0.378	1.381	-0.309	-0.269
	(0.049)	(0.050)	(0.044)	(0.058)
G_2	0.582	-0.859	0.050	-0.017
	(0.043)	(0.049)	(0.044)	(0.040)
G_3	0.118	-1.300	-0.146	-0.177
	(0.039)	(0.039)	(0.047)	(0.050)
G_4	0.749	-1.655	0.990	-0.031
	(0.046)	(0.048)	(0.045)	(0.050)
Z_2	-0.084	-0.141	0.346	-0.068
	(0.042)	(0.042)	(0.037)	(0.051)
Z_3	-1.047	0.020	0.461	-0.287
	(0.048)	(0.045)	(0.044)	(0.042)
Z_4	-0.123	0.108	-0.406	0.263
	(0.052)	(0.036)	(0.050)	(0.037)
Y_2	-1.069	0.230	0.057	-0.258
	(0.047)	(0.045)	(0.043)	(0.039)
Y_3	0.392	-0.292	1.403	-0.209
	(0.039)	(0.049)	(0.040)	(0.038)
Y_4	-0.200	-0.238	0.794	0.131
	(0.044)	(0.049)	(0.038)	(0.046)

Table S7: Transition costs high school

	Transition Cost HAVO
Z_2	70229.429
	(76.732)
Z_3	136365.636
	(99.518)
Z_4	147881.946
	(992.371)

Table S8: Distribution taste shocks

	EV Shocks
Scale	93225
	(90)

S3 Model fit

Table S9: Degree combinations by income

Income Quartile	Degree Combination	Observed	Estimated
0	havo	0.040	0.044
	<i>havo – bachelor</i>	0.117	0.130
	mbo3	0.129	0.137
	<i>mbo3 – mbo4</i>	0.091	0.089
	<i>mbo3 – mbo4 – bachelor</i>	0.030	0.034
	mbo4	0.279	0.267
	vmbo	0.124	0.106
1	havo	0.037	0.035
	<i>havo – bachelor</i>	0.113	0.119
	mbo3	0.135	0.124
	<i>mbo3 – mbo4</i>	0.096	0.100
	<i>mbo3 – mbo4 – bachelor</i>	0.031	0.032
	mbo4	0.287	0.304
	vmbo	0.085	0.097
2	havo	0.032	0.034
	<i>havo – bachelor</i>	0.131	0.117
	mbo3	0.121	0.123
	<i>mbo3 – mbo4</i>	0.089	0.084
	<i>mbo3 – mbo4 – bachelor</i>	0.030	0.039
	mbo4	0.284	0.297
	vmbo	0.075	0.071
3	havo	0.040	0.043
	<i>havo – bachelor</i>	0.164	0.166
	mbo3	0.105	0.107
	<i>mbo3 – mbo4</i>	0.077	0.072
	<i>mbo3 – mbo4 – bachelor</i>	0.031	0.034
	mbo4	0.267	0.268
	vmbo	0.077	0.066

Table S10: Degree combinations by academic potential

Prediction Quartile	Degree Combination	Observed	Estimated
0	havo	0.002	0.012
	<i>havo – bachelor</i>	0.011	0.019
	mbo3	0.210	0.178
	<i>mbo3 – mbo4</i>	0.121	0.143
	<i>mbo3 – mbo4 – bachelor</i>	0.027	0.036
	mbo4	0.325	0.336
	vmbo	0.145	0.125
1	havo	0.012	0.023
	<i>havo – bachelor</i>	0.037	0.050
	mbo3	0.146	0.134
	<i>mbo3 – mbo4</i>	0.107	0.096
	<i>mbo3 – mbo4 – bachelor</i>	0.036	0.038
	mbo4	0.326	0.342
	vmbo	0.105	0.094
2	havo	0.037	0.044
	<i>havo – bachelor</i>	0.125	0.121
	mbo3	0.093	0.115
	<i>mbo3 – mbo4</i>	0.083	0.076
	<i>mbo3 – mbo4 – bachelor</i>	0.038	0.036
	mbo4	0.290	0.287
	vmbo	0.072	0.080
3	havo	0.097	0.076
	<i>havo – bachelor</i>	0.342	0.334
	mbo3	0.047	0.069
	<i>mbo3 – mbo4</i>	0.043	0.033
	<i>mbo3 – mbo4 – bachelor</i>	0.023	0.030
	mbo4	0.181	0.175
	vmbo	0.047	0.047

Table S11: Program enrollment by academic potential and school type

Programme	School Type	Prediction Quartile	Observed	Estimated
havo	0	0	0.030	0.020
		1	0.050	0.055
		2	0.130	0.132
		3	0.361	0.374
	1	0	0.038	0.041
		1	0.084	0.099
		2	0.199	0.205
		3	0.481	0.475
	2	0	0.044	0.058

hbo	3	1	0.112	0.115
		2	0.256	0.242
		3	0.544	0.550
		0	0.067	0.087
	0	1	0.168	0.190
		2	0.374	0.359
		3	0.658	0.690
		0	0.292	0.350
	1	1	0.423	0.474
		2	0.527	0.527
		3	0.696	0.698
		0	0.320	0.377
	2	1	0.457	0.513
		2	0.570	0.571
		3	0.741	0.749
		0	0.325	0.362
	3	1	0.455	0.490
		2	0.579	0.563
		3	0.747	0.768
		0	0.339	0.361
mbo3	0	1	0.467	0.502
		2	0.597	0.591
		3	0.749	0.783
		0	0.520	0.487
	1	1	0.448	0.373
		2	0.329	0.344
		3	0.195	0.219
		0	0.510	0.472
	2	1	0.418	0.353
		2	0.315	0.300
		3	0.175	0.184
		0	0.489	0.489
	3	1	0.393	0.360
		2	0.281	0.303
		3	0.154	0.168
		0	0.471	0.478
mbo4	0	1	0.390	0.356
		2	0.273	0.283
		3	0.141	0.152
		0	0.827	0.866
	1	1	0.873	0.892
		2	0.856	0.844
		3	0.683	0.679
		0	0.839	0.867
	2	1	0.880	0.872
		2	0.826	0.803
		3	0.590	0.611
		0	0.843	0.845
	3	1	0.865	0.852

	2	0.789	0.768
	3	0.524	0.549
3	0	0.844	0.827
	1	0.835	0.803
	2	0.695	0.686
	3	0.441	0.444

Table S12: Enrollment proportions by income

Programme	Income Quartile	Observed	Estimated
havo	0	0.230	0.239
	1	0.209	0.209
	2	0.219	0.207
	3	0.260	0.284
hbo	0	0.487	0.530
	1	0.495	0.491
	2	0.530	0.558
	3	0.566	0.611
mbo3	0	0.363	0.365
	1	0.365	0.353
	2	0.336	0.318
	3	0.303	0.279
mbo4	0	0.759	0.749
	1	0.775	0.768
	2	0.766	0.788
	3	0.735	0.739

Table S13: Final schooling ages by school type

School Type	Prediction Quartile	Age Range	Observed	Estimated
0	0	0-5	0.630	0.633
		10-15	0.030	0.038
		5-10	0.340	0.329
	1	0-5	0.537	0.516
		10-15	0.046	0.052
		5-10	0.417	0.432
	2	0-5	0.471	0.468
		10-15	0.048	0.060
		5-10	0.481	0.471
	3	0-5	0.358	0.333
		10-15	0.067	0.064
		5-10	0.575	0.603
1	0	0-5	0.599	0.604
		10-15	0.029	0.039
		5-10	0.371	0.357
	1	0-5	0.513	0.492
		10-15	0.046	0.055
		5-10	0.442	0.453
	2	0-5	0.432	0.439
		10-15	0.059	0.065
		5-10	0.510	0.496
	3	0-5	0.336	0.303
		10-15	0.076	0.061
		5-10	0.588	0.636
2	0	0-5	0.596	0.627
		10-15	0.041	0.036
		5-10	0.363	0.338
	1	0-5	0.513	0.519
		10-15	0.049	0.052
		5-10	0.437	0.429
	2	0-5	0.425	0.457
		10-15	0.052	0.058
		5-10	0.523	0.486
	3	0-5	0.345	0.292
		10-15	0.073	0.055
		5-10	0.582	0.654
3	0	0-5	0.583	0.635
		10-15	0.040	0.034
		5-10	0.377	0.330
	1	0-5	0.510	0.513
		10-15	0.056	0.047
		5-10	0.434	0.440
	2	0-5	0.417	0.448
		10-15	0.061	0.053
		5-10	0.522	0.499
	3	0-5	0.337	0.295
		10-15	0.081	0.048
		5-10	0.581	0.657

Table S14: Final schooling ages by income

Income Quartile	Age Range	Observed	Estimated
0	0-5	0.485	0.483
	10-15	0.071	0.050
	5-10	0.445	0.467
1	0-5	0.483	0.520
	10-15	0.053	0.045
	5-10	0.464	0.435
2	0-5	0.470	0.461
	10-15	0.044	0.053
	5-10	0.486	0.487
3	0-5	0.453	0.413
	10-15	0.048	0.058
	5-10	0.500	0.529

Table S15: Wage equation no bachelor's degree

	Observed	Estimated
Coefficients		
Intercept	2.861	2.609
Experience	0.007	-0.000
Experience ²	0.001	-0.000
Income Quart. 2	0.006	-0.011
Income Quart. 3	0.016	0.017
Income Quart. 4	0.030	-0.004
mbo3	0.052	-0.013
Experience $\times$ mbo3	-0.006	-0.001
mbo4	0.049	0.051
Experience $\times$ mbo4	-0.002	0.001
Potential Quart. 2	0.008	0.018
Potential Quart. 3	0.011	-0.003
Potential Quart. 4	0.017	-0.002
RSE	0.207	0.210
vmbo	0.019	-0.049
Experience $\times$ vmbo	-0.009	-0.004

Table S16: Wage equation bachelor's degree holder

	Observed	Estimated
Coefficients		
Intercept	3.005	2.863
Experience	0.042	0.024
Experience ²	-0.002	-0.001
Income Quart. 2	-0.000	-0.022
Income Quart. 3	0.001	-0.000
Income Quart. 4	0.006	-0.016
<i>mbo3 – mbo4 – bachelor</i>	-0.024	-0.024
<i>mbo4 – bachelor</i>	0.002	-0.055
Potential Quart. 2	-0.002	0.019
Potential Quart. 3	-0.002	0.004
Potential Quart. 4	0.007	-0.015
RSE	0.193	0.200
Duration Uni	0.002	-0.002

Table S17: Wage equation bachelor's degree holder

	Observed	Estimated
Coefficient		
Intercept	3.024	2.987
Potential Quart. 2	0.025	0.034
Potential Quart. 3	0.039	0.024
Potential Quart. 4	0.061	0.047
School Type 2	-0.020	0.005
School Type 3	-0.013	-0.001
School Type 4	-0.016	-0.012

Table S18: Wage equation bachelor's degree holder

	Observed	Estimated
Coefficient		
Intercept	3.127	3.077
Potential Quart. 2	0.005	0.023
Potential Quart. 3	0.004	0.012
Potential Quart. 4	0.021	0.017
School Type 2	-0.022	-0.003
School Type 3	-0.019	-0.006
School Type 4	-0.021	-0.018

Table S19: Wage equation bachelor's degree holder

	Observed	Estimated
Coefficient		
Intercept	3.053	3.059
Potential Quart. 2	0.029	0.044
Potential Quart. 3	0.037	0.040
Potential Quart. 4	0.066	0.051
School Type 2	-0.019	-0.012
School Type 3	-0.018	-0.036
School Type 4	-0.023	-0.020

Table S20: Wage equation bachelor's degree holder

	Observed	Estimated
Coefficient		
Intercept	0.580	0.740
Potential Quart. 2	0.108	0.064
Potential Quart. 3	0.193	0.099
Potential Quart. 4	0.283	0.147
School Type 2	0.031	0.002
School Type 3	0.016	-0.001
School Type 4	-0.007	-0.001

Table S21: Wage equation bachelor's degree holder

	Observed	Estimated
Coefficient		
Intercept	0.336	0.452
Potential Quart. 2	0.084	0.088
Potential Quart. 3	0.202	0.157
Potential Quart. 4	0.287	0.265
School Type 2	0.020	-0.002
School Type 3	0.018	0.001
School Type 4	0.008	0.001

Table S22: Wage equation bachelor's degree holder

School Type	Prediction Quartile	Observed	Estimated
0	0	0.292	0.343
	1	0.415	0.454
	2	0.501	0.479
	3	0.609	0.587
1	0	0.317	0.361
	1	0.442	0.481
	2	0.525	0.504
	3	0.615	0.620
2	0	0.320	0.341
	1	0.434	0.450
	2	0.523	0.474
	3	0.607	0.620
3	0	0.328	0.324
	1	0.435	0.432
	2	0.507	0.451
	3	0.578	0.558

Table S23: Wage equation bachelor's degree holder

School Type	Prediction Quartile	Observed	Estimated
0	0	0.157	0.194
	1	0.262	0.281
	2	0.350	0.334
	3	0.469	0.460
1	0	0.180	0.207
	1	0.273	0.298
	2	0.352	0.347
	3	0.461	0.481
2	0	0.188	0.193
	1	0.261	0.281
	2	0.361	0.329
	3	0.441	0.481
3	0	0.180	0.179
	1	0.270	0.266
	2	0.352	0.312
	3	0.431	0.435